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01CE0717 – DevOps – Lab Manual

A.Y. 2026-27

DevOps Essentials (01CE0717)

Lab Manual 2026-27

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Lab Program

Signature Marks

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Build a branching model to help your team understand the Git workflow for faster integration of work.

2

Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.

3 Create Jenkins job and pipeline it from GitHub repo to run using build.

4

Building a CI/CD Pipeline with Jenkins.

5

Dockerizing Jenkins Pipeline.

6

Deploy web app in AWS EC2 using Ansible.

7

Deploy Angular/React/Java/Python or any other Application in Docker Container

8

Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.

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Practical 1: Build a branching model to help your team understand the Git workflow for faster integration of work.

Step 1: Initialize Git Repository

```
mkdir git-branching-demo  
cd git-branching-demo  
git init
```

Step 2: Initial Commit

```
echo "# Git Branching Demo" > README.md  
git add .  
git commit -m "Initial commit"
```

Step 3: Create Develop Branch

```
git branch develop  
git checkout develop  
git branch
```

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Step 4: Create Feature Branch

```
git checkout -b feature/login
```

Step 5: Work on Feature

```
echo "Login feature code" > login.txt  
git add .  
git commit -m "Added login feature"
```

Step 6: Merge Feature into Develop

```
git checkout develop  
git merge feature/login
```

Step 7: Create Release Branch

```
git checkout -b release/v1.0
```

Step 8: Merge into Main

```
git checkout main
```

```
git merge release/v1.0
```

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Step 9: Hotfix Branch

```
git checkout -b hotfix/bugfix
```

Step 10: Merge Hotfix

```
git checkout main
```

```
git merge hotfix/bugfix
```

```
git checkout develop
```

```
git merge hotfix/bugfix
```

Step 11: Branch Graph

```
git log --oneline --graph --all
```

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Practical 2: Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.

Step 1: Install Java

Download JDK from: [Oracle JDK Downloads](#)

1. Welcome Screen : The JDK Setup Wizard opens. Click Next.

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2. Choose Installation Folder : Leave the default installation path (recommended) or choose another folder,
then click Next.

3. Installation in Progress : Wait while the installer copies the required files.

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4. Installation Complete: Click Close to finish.

Configure JAVA_HOME

1. Open System Properties !' Advanced System Settings.
2. Click Environment Variables.

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3. Under System Variables, click New.

(Variable Name: JAVA_HOME

Variable Value: C:\Program Files\Java\jdk-21)

4. Edit Path and add:
%JAVA_HOME%\bin

Verify Installation

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Step 2: Install Apache Maven
Download from: [Apache Maven Official Website](https://maven.apache.org/)

Extract Maven: Extract the ZIP file to:(C:\Program Files\)

Configure MAVEN_HOME

1. Open Environment Variables.
2. Create a new system variable:
Variable Name: MAVEN_HOME
Variable Value: C:\Program Files\Apache\maven

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3. Add to Path:
%MAVEN_HOME%\bin

Verify Maven Installation

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Step 3: Install Gradle

Download Gradle: [Gradle Official Website](https://gradle.org/)

Extract Gradle: Extract to (C:\Gradle\gradle-9.x)

Configure GRADLE_HOME

1. Create a new system variable:(

Variable Name: GRADLE_HOME

Variable Value: C:\Gradle\gradle-9.x)

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2. Add to Path:

%GRADLE_HOME%\bin

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Verify Gradle Installation

Step 4: Create a Maven Project

mvn archetype:generate

-DgroupId=com.example
-DartifactId=MavenDemo
-DarchetypeArtifactId=maven-archetype-quickstart
-DinteractiveMode=false

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Move to project directory: `cd MavenDemo`

Build Project: `mvn clean install`

Run: `java -cp target\MavenDemo-1.0-SNAPSHOT.jar com.example.App`

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Step 5: Create a Gradle Project Create
project: `gradle init`
Choose: Application > Java > Groovy DSL

Build project: `gradle build`

Run application: `gradle run`

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Practical-3. Create Jenkins job and pipeline it from GitHub repo to run using build.

1. Install Jenkins in your system.

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2. Now login to your Jenkins server site at <http://localhost:8080>

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